



User Guide

MDKE9 Smart Operating Panel



User Guide

Preface

Thank you for purchasing the MDKE9 series smart operating panel developed by Suzhou Inovance Technology CO., Ltd.

MDKE9 smart operating panel is a new generation AC drive commissioning assistant for single drives, such as MD200 series, MD310 series, MD500 series, MD500E series, and MD290 series. Featuring LCD display, parameter setting, status monitoring, parameter copy, fault analysis, program downloading, USB mass storage functions.

This user guide describes the correct use of the MDKE9 series smart operating panel, including safety precautions, product information, installation, and wiring, function and application, maintenance and repair. Read and understand the user guide before use. Contact our customer service center if you have any question during the use.



Safety Instructions

The drawings in the user guide are sometimes shown without covers or protective guards. Remember to install the covers or protective guards as specified first, and then perform operations in accordance with the instructions.

The drawings in the user guide are shown for description only and may not match the product you purchased.

The instructions are subject to change, without notice, due to product up-grade, specification modification as well as efforts to increase the accuracy and convenience of the user guide.

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Chapter 1 Safety Instructions

1.1 Safety Disclaimer

- 1) Read and comply with the safety instructions during installation, operation, and maintenance on the equipment.
- 2) To ensure the safety of humans and the products, follow the marks on the products and all the safety instructions in this document.
- 3) The “CAUTION”, “WARNING”, and “DANGER” are only supplements to the safety instructions.
- 4) Use this equipment based on the designated environment requirements. Damages caused by improper usage are not covered by warranty.
- 5) The company shall take no responsibility of any personal injuries or property damages caused by improper usage.

1.2 Safety Levels and Definitions

The  **Danger** “Danger” sign indicates that failure to comply with the notice will result in server personal injuries or even death.

The  **Warning** “Warning” sign indicates that failure to comply with the notice may result in server personal injuries or even death.

The  **Caution** “Caution” sign indicates that failure to comply with the notice may result in minor or moderate personal injuries or damage to the equipment.

1.3 Safety Instructions

Unpacking	
 Caution	◆ Check whether the packing is intact and whether there is damage, water seepage, damp, and deformation. ◆ Unpack the package following the package sequence. Do not hit the package with force. ◆ Check whether there are damages, rust, or injuries on the surface of the product or product accessories. ◆ Check whether the number of packing materials is consistent with the packing list.
 Warning	◆ Do not install the equipment if you find damages, corrosions, or indications of use on the product or accessories. ◆ Do not install the equipment if you find water seepage, component missing or damage upon unpacking. ◆ Do not install the equipment if you find the packing list does not conform to the equipment you received.

Storage and Transportation

 Caution

- ◆ Store this equipment based on the storage and transportation requirements on humidity and temperature.
- ◆ Avoid transporting the equipment in environment such as water splashing, rain, direct sunlight, high voltage, strong magnetic field, and strong vibration.
- ◆ Avoid storing the product for more than 3 months, long-term storage shall require stricter protection and necessary inspections..
- ◆ Pack the product strictly before you transport. Use a sealed box for long-distance transportation.
- ◆ Never transport this product with products or materials that harm or have negative impacts on this product.

 Warning

- ◆ Use professional loading and unloading equipment to carry large-scale or heavy products.
- ◆ When carry this equipment with bear hand, hold the product casing firmly with care preventing from parts falling, otherwise, it may result in personal jury or equipment damages.
- ◆ Handle the products with care, mind your steps, otherwise, it may result in personal juries or equipment damages.
- ◆ Never stand or stay below the equipment when the equipment is lifted by hoisting equipment.

Installation

 Warning

- ◆ Read and comply with the safety instructions before performing installation.
- ◆ Do not modify this product.
- ◆ Do not rotate the product components or the fixed bolts and red marked bolts on product components.
- ◆ Do not install this product in places where there is strong electric field or strong magnetic field.
- ◆ When this product is installed in a cabinet or terminal device, the cabinet or terminal device must be equipped with protective shell. The proof class must comply with relevant IEC standards and local regulations.

 Danger

- ◆ Do not allow non-professionals to perform product installation, wiring, maintenance, inspection or parts replacement.
- ◆ The installation, wiring, maintenance, inspection and parts replacement are intended to be performed by professional personnel only.
- ◆ Installation personnel must be familiar with product installation requirements and relevant technical materials.
- ◆ If you need to install transformer or other strong electromagnetic interference equipment, install shielding device at the same time to avoid product malfunction.

Wiring

**Danger**

- ◆ Do not allow non-professionals to perform product installation, wiring, maintenance, inspection or parts replacement.
- ◆ Never perform wiring at power-on. Failure to comply may result in electric shock.
- ◆ Cut off all power supplies before wiring. Wait at least 10 minutes after power-off so that residual voltage can discharge safely. Wait at least 10 minutes after power-off so that residual voltage can discharge safely.
- ◆ Make sure that the equipment is reliably grounded. Failure to comply may result in electric shock.
- ◆ Follow the proper electrostatic discharge (ESD) procedures, and wear an anti-static wrist strap to perform wiring. Failure to comply may result in damage to the equipment or the internal circuit the product.

**Warning**

- ◆ Never connect the power cable with the product output terminals (U, V, W.). Failure to comply may result in equipment damage or even fire.
- ◆ When connecting driving equipment with the motor, make sure the phase sequence of the drive and motor are consistent to prevent motor reverse rotation.
- ◆ The cable used in wiring must conform to the wire diameter and shielding requirements, the shielding layer of the shielded cable must be reliably grounded at one end.
- ◆ After wiring, make sure there are no fallen screws or exposed wire inside the cabinet or product.

1

Power-on

**Danger**

- ◆ Before power-on, make sure that the equipment or products are intact, all wiring are safely connected, and the motor device allows a restart.
- ◆ Before power-on, check if the power supply meets the equipment requirements, avoid equipment damage or fire.
- ◆ At power-on, equipment or product may start running abruptly, keep away from mechanical device.
- ◆ After power-on, never open the cabinet door or the protective shell to avoid electric shock.
- ◆ Never touch any terminals at power-on to avoid electric shock.
- ◆ Never dismantle the equipment or remove any parts from the equipment at power-on to avoid electric shock.

Operation

**Danger**

- ◆ Never touch any terminals during operation. Failure to comply may result in electric shock.
- ◆ Never dismantle the equipment or remove any parts from the equipment during operation. Failure to comply may result in electric shock.
- ◆ Never touch the equipment shell, fan or resistor for temperature detection. Failure to comply may result in burn!
- ◆ Do not allow non-professional technicians to detect signals during operation. Failure to comply may result in equipment damage even personal injuries.

Warning

- ◆ Prevent metal or other objects from falling into the device during operation, failure to comply may result in equipment damage.
- ◆ Never use contactors to start or stop the equipment. Failure to comply may result in damage to the equipment!

Maintenance

Danger

- ◆ Do not allow non-professionals to perform product installation, wiring, maintenance, inspection or parts replacement.
- ◆ Never perform maintenance at power-on. Failure to comply may result in electric shock.
- ◆ Before maintenance, wait at least 10 minutes after power-off so that residual voltage can discharge safely.

Warning

- ◆ Following the equipment repair and maintenance requirements to perform routine and periodical inspection and maintenance, and keep a maintenance record.

Repair

Danger

- ◆ Do not allow non-professionals to perform product installation, wiring, maintenance, inspection or parts replacement.
- ◆ Never perform any inspection or maintenance operations at power-on. Failure to comply may result in electric shock.
- ◆ Before inspection or maintenance, wait at least 10 minutes so that residual voltage can discharge safely.

Warning

- ◆ Follow the terms defined in the warranty agreement to repair the product.
- ◆ When the equipment has a fault or is damaged, follow the instructions of the professionals to perform troubleshooting or maintenance, and keep a maintenance record.
- ◆ Following the quick-wear parts instructions to replace the damaged parts.
- ◆ Do not continue to use the damaged equipment. Failure to comply may result in worse damages.
- ◆ Following the equipment repair and maintenance requirements to perform routine and periodical inspection and maintenance, and keep a maintenance record.

Disposal

Warning

- ◆ Following the local regulations or standards to dispose the retired equipment or products. Failure to comply may result in property damage or even death.
- ◆ Following the industry waste disposal standards to recycle the scrapped equipment to avoid environmental pollution.

1.4 Safety Signs

To guarantee safety operations and maintenance of the equipment, follow the safety stickers on the equipment and products. Do not stain or remove the safety signs. Instructions of safety signs are as follows:

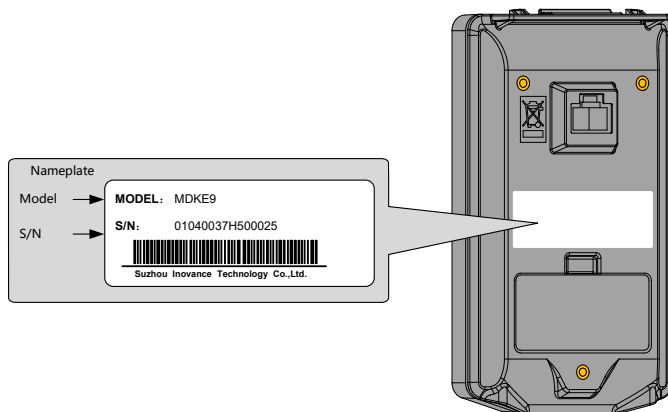
Safety signs	Instructions
	◆ Read this notice before installation and operation. Failure to comply may result in electric shock. ◆ Do not remove the cover within 10 minutes of power-off. ◆ During maintenance, inspection or wiring, you may start operation at least 10 minutes after power-off at the input/output terminal when the power indicators are completely off.

Chapter 2 Product Information

MDKE9 smart operating panel is a new generation AC drive commissioning assistant for single drives, such as MD200 series, MD310 series, MD500 series, MD500E series, and MD290 series.

2.1 Nameplate

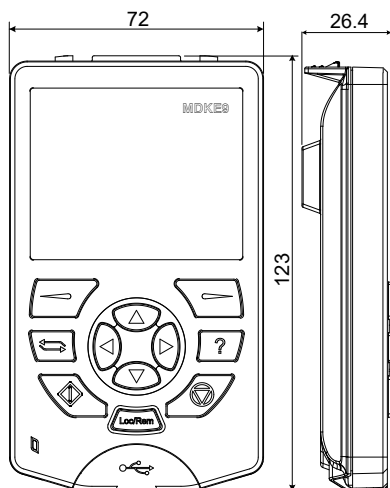
Figure 2-1 Nameplate



2

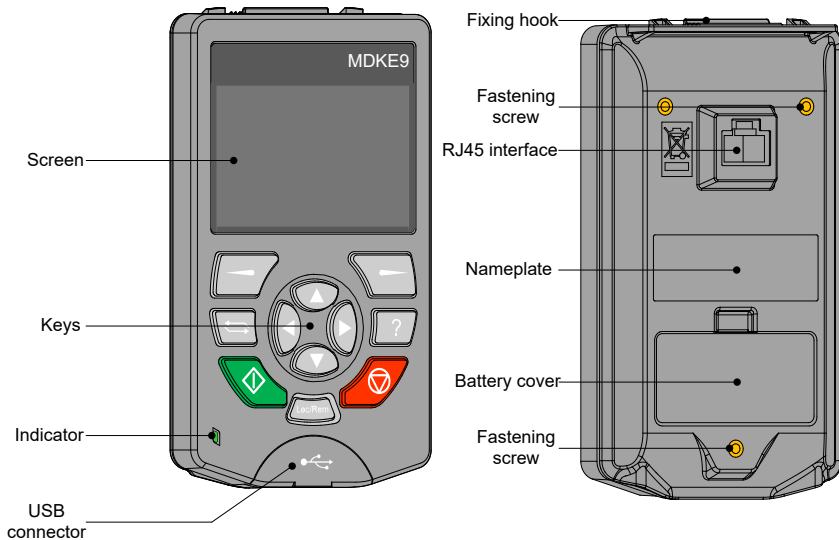
2.2 MDKE9 Dimensions

Figure 2-2 MDKE9 dimensions (Unit: mm)



2.3 Components

Figure 2-3 MDKE9 Components



- USB connector

Connect the MDKE9 to a PC via a USB cable for data exchange after connection, it can work in USB disk mode to read/write the MDKE9 SD card through PC.

- RJ45 connection terminal

To connect MDKE9 to the AC drive. Mechanical connection is completed through the top buckle.

- Nameplate

To show information such as product model and serial number

- Battery cover

The battery and SD card slot are beneath the cover. The battery provides power to the real-time clock. The SD card stores factory parameters and real-time data of the AC drive, which helps in fault diagnosis and analysis. The SD card can also be used to store user program, update firmware or update device.

- Fixing hook

To fix the MDKE9 to the base

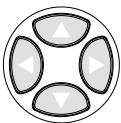
- Fastening screw

The MDKE9 can be installed on the cabinet door by using the installation package (IP20). When the system protection class is high (IP54), you may select the installation package with higher protection class (MDKE9 panel with IP54 installation package), and then fix the MDKE9 with the fastening screws.

2.4 Key Introduction

2.4.1 Keys and Functions

Table 2-1 Keys and functions

Key	Name	Function
	Left	To return or cancel.
	Right	To select or confirm.
	Shift	To quickly enter the device list On some interfaces, you are not allowed to switch over the devices and the key is invalid.
	Help	To open the help interface The content on the help page relevant to corresponding menu or interface
	Run	To start the device in local control mode
	Stop	To stop the device in local control mode To reset the fault when the device in fault status
	Loc/Rem Switchover key	To switch between the local control, remote control, and terminal control mode
	Direction	To use  and  to select items in the menu and list, scroll the text page, and adjust the values (time setting, password entering, or parameter value modifying). To use  and  to move the cursor left or right when you are modifying parameters.

2.4.2 Shortcut

Table 2-2 Shortcut introduction

Shortcut	Scope	Function
Left+Up	Any view	To adjust the backlight brightness
Left+Down		
Right+Up	Any view	To adjust the contrast
Right+Down		
Left+Right	Parameter list view	Add the function code where is cursor is to shortcut interface
Left+Right	Shortcut view	Delete the function code from the shortcut commissioning interface
Left+Help	Function code editing view	Recover to the original value

2.5 Indicator Status

The status indicator is used to indicate whether a fault or alarm exists.

Table 2-3 Indicator status status

Indicator state	Demonstration	Meaning
Steady on in green		The device runs normally.
Flashing in green		Data is transferring between the MDKE9 and PC via USB.
Steady on in red		Communication between the MDKE9 and the device is normal, but the device is faulty.
Flashing in red		Communication between the MDKE9 and the device is interrupted.

2.6 General Specifications

Table 2-4 General Specifications

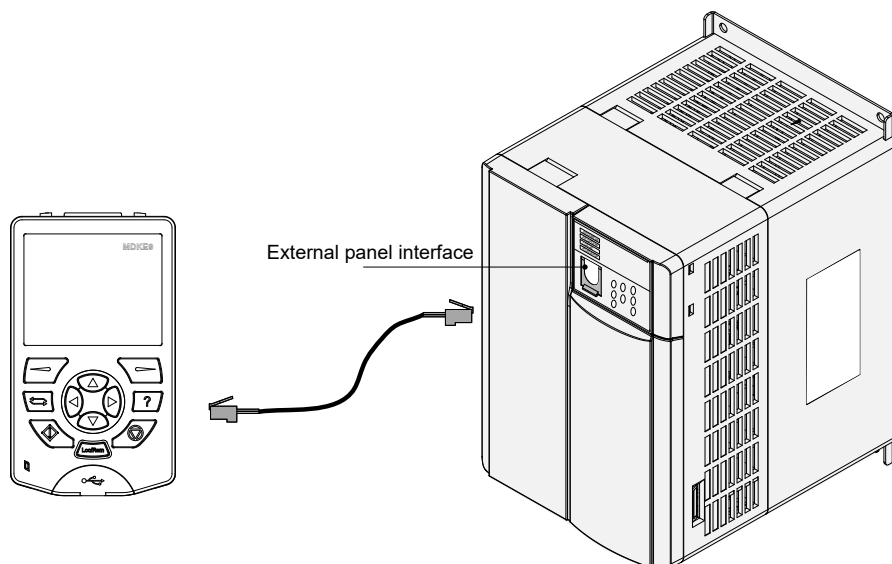
S.N	Item	Running	Storage	Transportation
1	Working temperature range	-20 ℃ ~+55 ℃	-40 ℃ ~+70 ℃	-40 ℃ ~+70 ℃
2	Attitude requirement	< 4000 m	-	-
3	Relative humidity	95%(non-condensing)		
4	Pollution / class	3C3(IEC 60721-3-3, IEC 60721-3-2, IEC 60721-1-3-1)		
5	Vibration	61800-5-1 ed 2 EN 60082-2-6 test Fc (1g)	-	2M3 class (EN 60082-2-6)
6	impact	3M4 class (EN 60062-2-27)	-	2M2 class (EN 60082-2-27)
7	Free drop	FN -60068-2-32 Drop height 1 m	-	-
8	IP class	Connect to drive equipment: IP54 (select IP54 accessories) Single unit: IP20		
9	Electromagnetic compatibility	IEC 61800-3		
10	Material	Disposal: The MDKE9 is recyclable. Refer to relevant international and local laws and regulations for disposal and recycling.		

Chapter 3 Installation and Wiring

The MDKE9 is suitable for multiple Inovance's AC drives. This user guide takes M310 as the example to describe the installation and wiring of the MDKE9. Refer to the corresponding user guide for details if the MDKE9 is installed on other AC drives.

3.1 Connect the MDKE9 with AC Drive

Open the external panel cover on MD310 AC drive, connect the RJ45 connection terminal at the back of the MDKE 9 via a network cable (optional).



3

Table 3-1 Recommended Inovance standard network cable

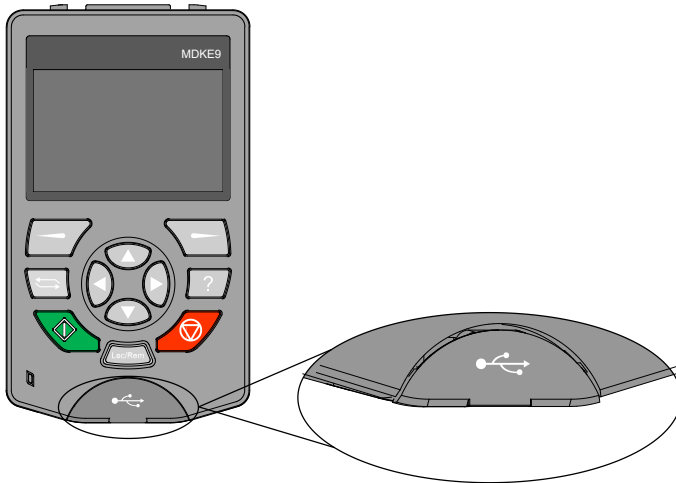
Purchase code	Model	Description
01040020	C45590-GNCN-C25003	Standard network cable with length of 3000 mm
01040017	C45590-GNCN-C250015	Standard network cable with length of 1500 mm
01040016	C45590-GNCN-C2500080	Standard network cable with length of 800 mm

3.2 Connect MDKE9 with PC

The mini-B type USB connector of the MDKE9 is to connect the PC, the USB cable length cannot exceed 3 m. The communication protocol is USB2.0 between the MDKE9 and the PC.

- 1) Open the cover at the USB connection terminal;

Figure 3-1 Connect MDKE9 with PC (step 1)



- 2) Plug in the USB cable, suggest to use cable with Ferrite toroidal, then connect the cable to PC.

Figure 3-2 Connect MDKE9 with PC (step 2)

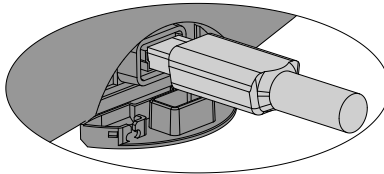
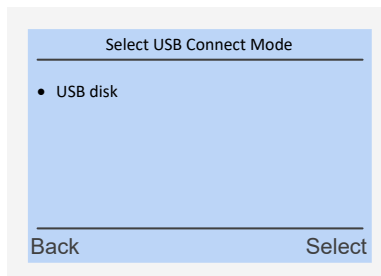


Figure 3-3 MDKE9 display after connecting to PC.



Note

After selecting USB mass storage mode, the MDKE9 can be used as storage device.

3.3 Install MDKE9 on the Cabinet Door

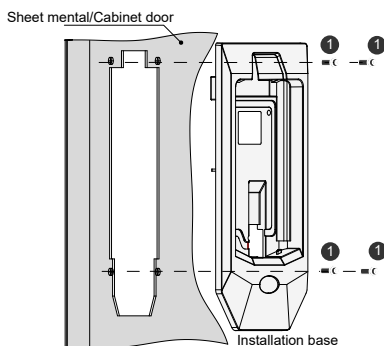
When installing the MDKE9 panel on the cabinet door, install the panel base first (optional), the installation procedures are as follows:

Purchase code	Model	Description
01040022	CP600-BASE1	Panel base installation package

1) Panel base installation

Use the four screws **1** as shown in figure 3-4 to fix the base on the cabinet door.

Figure 3-4 Installing base on the cabinet door

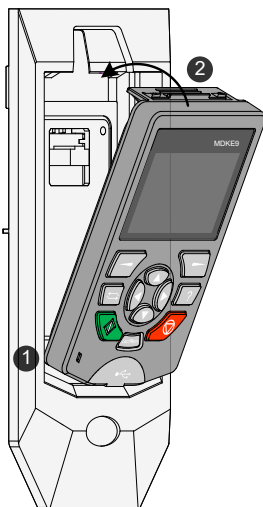


2) Installing MDKE9.

Place the bottom of the MDKE9 keyboard at the bottom of the base slot **1**. Rotate the MDKE9 **1** at the rotation point and push the upper part **2** until the click click, and the keyboard is mounted on the base successfully.

3

Figure 3-5 Installing the MDKE9 into the base

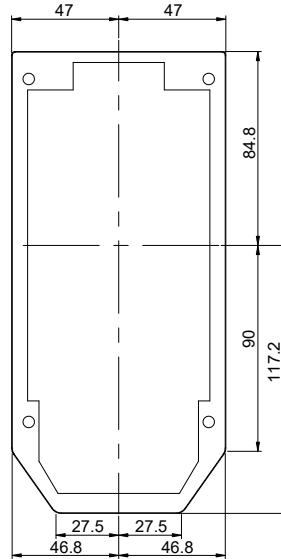
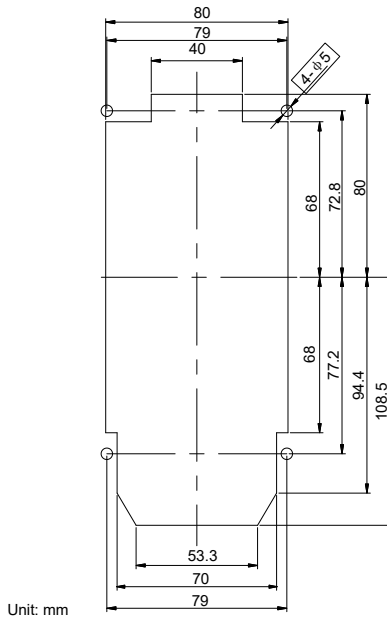


- 3) Take the MDKE9 out.

Press clip ② in figure 3-5 with the index finger and middle finger to release the MDKE9, then pull the upper end of the MDKE9 out of the base slot.

Figure 3-6 Sheet metal slot dimensions

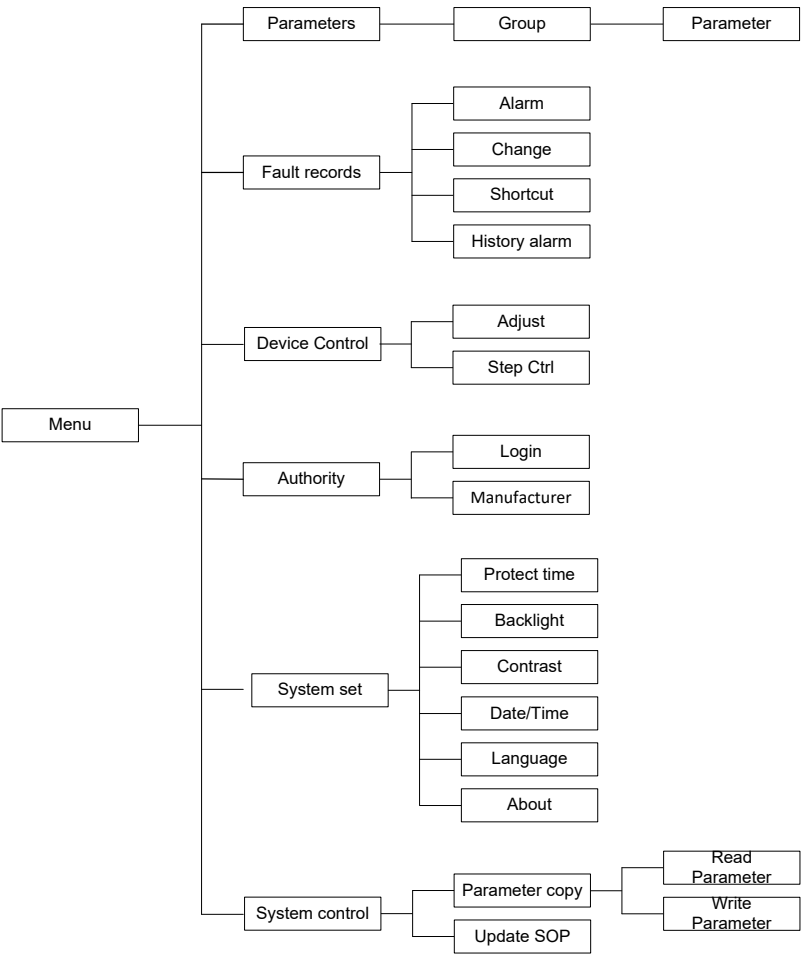
Figure 3-7 Base mounting hole size



Chapter 4 Functions and Applications

4.1 Summary

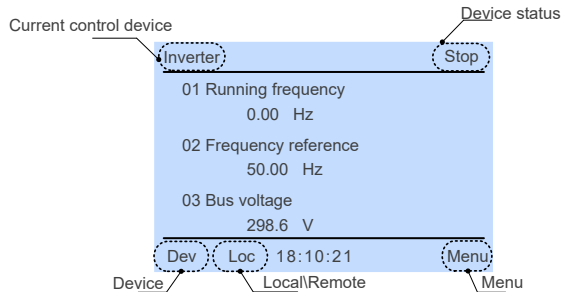
Figure 4-1 MDKE9 menu structure diagram



4.1.1 Main Interface

Main interface: used to monitor signals. The interface is as below:

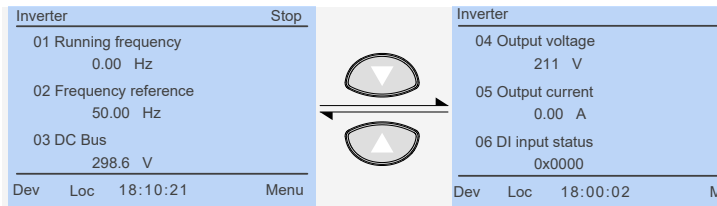
Figure 4-2 Main interface



- Current device name: to display the name of the device currently controlled by the MDKE9
- Device status/fault information: to display the current running status of the device. When the device fails, the device shall flash display the fault information, running state shall not be displayed then.
- Device list: List of devices connected to the network
- Menu: to enter the function menus of the MDKE9

At main interface, you can monitor the monitoring equipment status, such as speed, torque, power. The main interface has one or more pages, and each page can display up to three signals by turning up and down the key.

Figure 4-3 Main interface view method

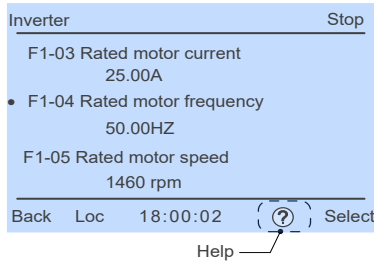


Note

Power-on default display "figure 4-2 main interface". At other pages, the device shall automatically return to the home page view display if there is no operating for more than 10 minutes.

4.1.2 Help

Figure 4-4 Help icon display demonstration



In some particular menus and interfaces, you can press  to open the corresponding help pages. The right corner of the screen shall display " ? " when there is available help information.

4.1.3 Base Operation

Table 4-1 Equipment operating instruction

Task	Action
Start and stop the equipment	Press  to start the equipment at local control mode Press  key to stop the equipment.
Switchover between local/remote/terminal control	Press  to switch between the local control, remote control, and terminal control mode.

4.1.4 Fault Viewing

Table 4-2 Fault description

Task	Action
Check the current faults	Open "menu"->"status monitoring"->"fault status"
Open the help information of the fault	Press  at the main interface.
Reset the fault of the current operating device.	Press 
to view the history fault information.	Open "menu"->"status monitoring"->"fault status".

4.1.5 Switching Operating Device

In the non-parametric editing interface, press  on the main interface or press  , you can enter the list of on line equipment page. The device list interface lists the devices and their status information in the current system. After  ,  selecting the equipment to be operate on via "direction key" , press  , then wait until the keyboard operation completes. When the switch is complete, the upper left corner of the page shall display specific information about the current device.

4.1.6 Connecting to the PC

The MDKE9 is connected to the PC via a USB cable: USB mass storage mode You cannot operate the MDKE9 when it is connected to a PC.

Connecting steps:

- 1) Use one end of the USB cable to plug into the MDKE9 USB interface, the other end connect to USB interface of PC
- 2) In the pop-up connection selection interface, select the connection mode you need:

USB mass storage mode: By now the new removable disk in the PC is the content of SD card, which can be read and write.

4.2 Parameter Setting

Table 4-3 Parameter setting

Task	Action
View/editing parameter	Open "menu"->"parameter setting", select the parameter group to be modified, then select the parameter to be modified.
Recover to the original parameter value	After entering the editing mode, press  and  at the same time to restore to the original setting, then press  to save.
Adding parameters to "shortcut" interface	Add the function code to the shortcut interface On the main interface, enter any random function group, move the cursor to the required function code, and press  and the  at the same time to add the function code to the shortcut interface. Delete the function code from the shortcut interface On the fasting commissioning interface, move the cursor to the fuction code to be deleted, and press  /  at the same time to delete the function code from the fasting commissioning page.
View shortcut parameters	Open "menu"->"status monitoring"->"shortcut".
View the modified parameter settings	Open "menu"->"status monitoring"->"modified parameter".

4

4.2.1 Parameter Viewing and Modifying

You can view and modify parameters in parameter interface. The displayed parameters will be different under different control authority. (Refer to "4.5 Authority Management")

If a parameter group is selected, all the parameters under this group  will be displayed. You can view or modify these parameters.and press  to view the available help information.

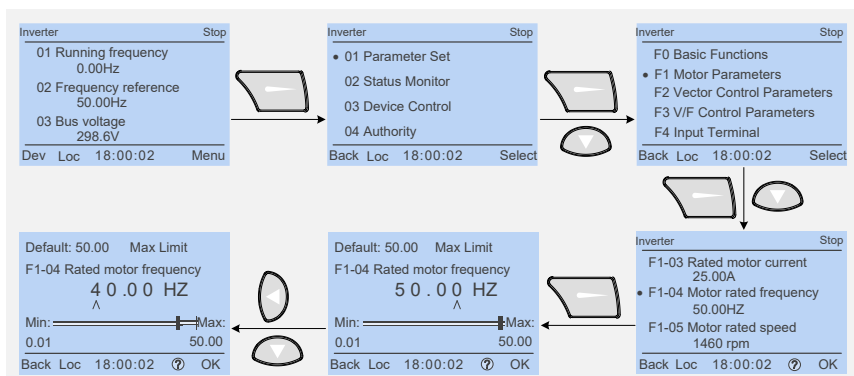
For read-only parameters, "Select" will not be displayed at the right bottom on the screen and pressing  has no effect.

If "0x" is displayed before the parameter, the parameter is a hexadecimal value.

1) Modifying digit parameters

to set F1-04 to 40.00Hz, that is taking the rates frequency of 40.00 Hz as an example. The setting steps are as follows:

Figure 4-5 Example of modifying digit parameters



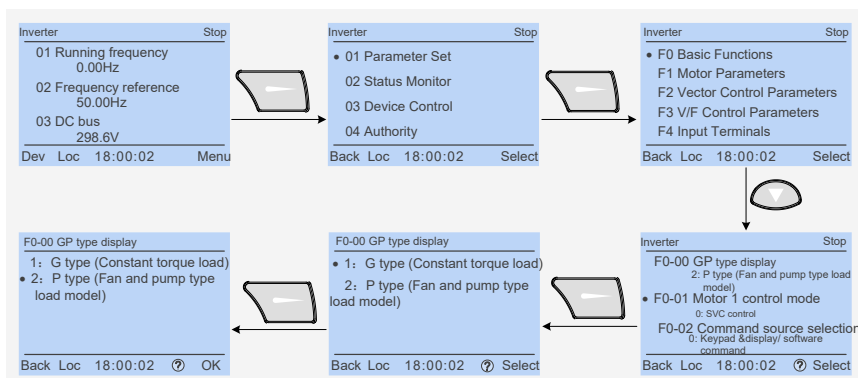
When the F1-04 is set to 40.00 Hz, press to save the setting and exit or press to quit and exit.

Press and at the same time to restore the parameter to default value, and then press to save the modification.

2) Modifying select list parameters

Take setting F0-00 to 2, that is to set GP type to P type as the example. The setting steps are as follows:

Figure 4-6 Modifying selection table parameter



When the F1-04 is set to 40.00 Hz, press to save the setting and exit or press to quit and exit.

Press and at the same time for 1.5s to restore the parameter to default value and then press to save the modification.

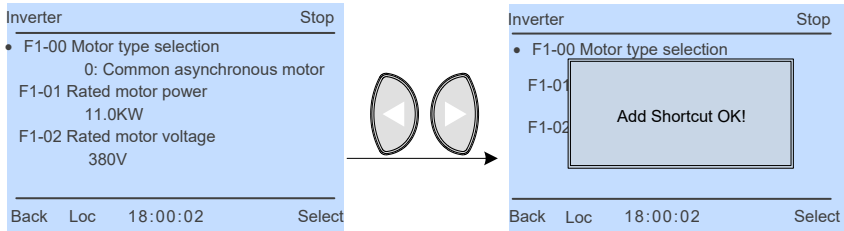
4.2.2 Add the Parameter to the "Shortcut" Interface

In parameter list, move the cursor to the parameter to be added and hold down the  and  for 1.5s. you can add 20 parameters to the shortcut interface and a prompt will appear to indicate whether the parameter is added successfully or not.

The parameters in the shortcut interface are in chronological order based on the time when they are added.

Take adding F1-00 to the shortcut interface as the example. The adding steps are as follows:

Figure 4-7 Example of adding parameters to the shortcut interface



On the shortcut interface, move the cursor to the function code to be deleted, and press  and  at the same time to delete the function code from the short cut interface

4.3 Status Monitoring

On status monitoring interface, you can view the current status of the fault equipment, the modified parameters, the shortcut, and the history fault.

Table 4-4 Status monitoring description

Task	Action
Check the current faults of all equipment	Open "menu"-"status monitoring"-"fault status"
View the modified parameters	Open "menu"-"status monitoring"-"modified parameter".
View the history fault information.	Open "menu"-"status monitoring"-"fault status". Delete history faults Press  after logging into the system. An interface pops up with selection "Delete history faults?" Press  again to complete the procedure.

4.3.1 Fault Viewing

In the fault interface, you can view all current faults of the device. Including the fault occurring time and detailed fault description.

4.3.2 Modified Parameters

Parameters that are different from the default values are listed in this page, and the order is determined by parameter numbers. Parameter scanning time may vary depending on the specific device, wait patiently based on the prompts.

4.3.3 "Shortcut" Interface

All user-defined parameters are listed on shortcut page, for detailed adding procedure, see "4.2.2 Adding Parameters to the "Shortcut" interface. On the shortcut interface, move the cursor to the function code to be

deleted, hold down  and  for 1.5s to delete the function code from the short cut interface.

4.3.4 History Faults

In the fault records interface, you can view the fault occurring time and detailed fault description.

Delete history faults

After logging into the system, pressing  on the "history faults" page, then press  again on the pop-up warning window. For details about how to logging into the system, refer to "Authority Management".

Note

History faults are stored on the SD card, make sure that you have correctly installed the SD card.

4.4 Equipment Operating

You can perform commissioning on equipment operating menu.

Table 4-5 Equipment operating

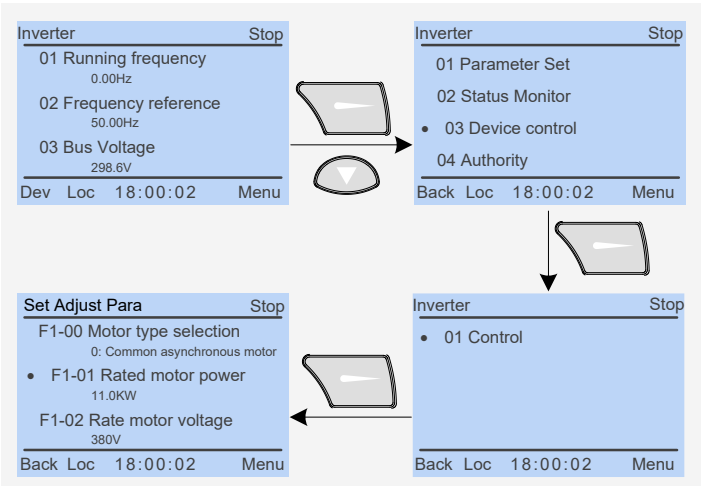
Task	Action
Start equipment commissioning	Open "menu"-"equipment operating"-" commission", press  "Run" to start commission.

4.4.1 Commission

This page lists the parameters to be set for equipment commissioning, after the setting is completed,

press  "run" to start commissioning.

Figure 4-8 Commissioning



4.5 Authority

You can login the equipment via panel operating on Authority Management interface.

4.5.1 Login

Table 4-6 Login description

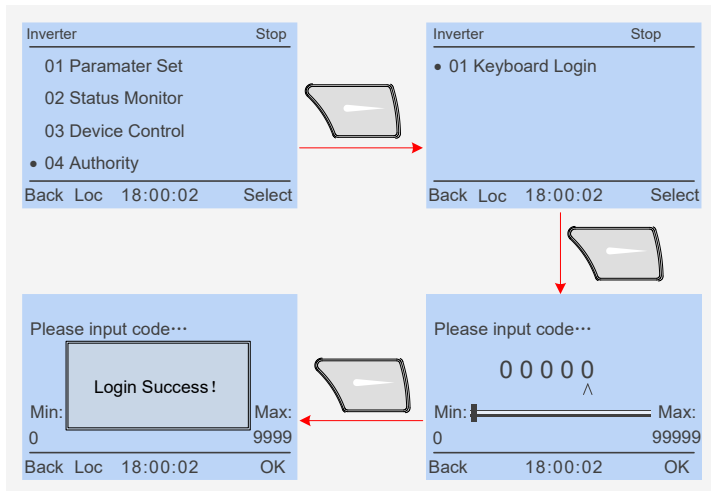
Task	Action
Logging into the system	Open "menu"-"authority management"-"logging on"

You have to log in to the MDKE9 before using the following functions under system control:

- Parameter copy: Both read parameter and write parameter, to backup and set all the device parameters quickly
- Device program burning: To update the device program via the MDKE9

Program update: To update the program of the MDKE9, enter the manufacturer password to log in to the MDKE9.

Figure 4-9 Example of login



4

4.6 System Setting

In system setting interface, you can set the protect time, date/time, language and view the system information.

Table 4-7 Description of system setting

Task	Action
Set protect time	Open "menu"-"system setting"-"protect time"
Adjust the backlight brightness	Press + / key
To adjust the contrast	Press + / key

Task	Action
Change the audio setting	Open "menu"-"system setting"-"language selection"
Change time	Open "menu"-"system setting"-"date/time"
Set backlight light off time	Open "menu"-"system setting"-"protect time"
Check panel version	Open "menu"-"system setting"-"about"

1) Set Protection time

The backlight might go off when there is no operation for a period of time. The unit of time is minute, support up to 99 minutes protect time setting. When it is set to 0, turn off the protect time function. When the panel automatically turns off protect time, pressing any key shall turn on the backlight again.

Note	After the backlight automatically turns off, if another fault occurs, the backlight shall be on again, after
-------------	--

2) Date/Time set

Pressing  +  to select the time/date to be modified, press  to save.

3) Language selection

It decides the language of the interface (Chinese or English).

4) About the system

To view the version information of the MDKE9.

4.7 System Operation

You have to log in the MDKE9 before using the functions such as parameter copy, update device, update MDKE9 under system control. Refer to section 4.5.1 for details. For details of login method, see "4.5.1 Login".

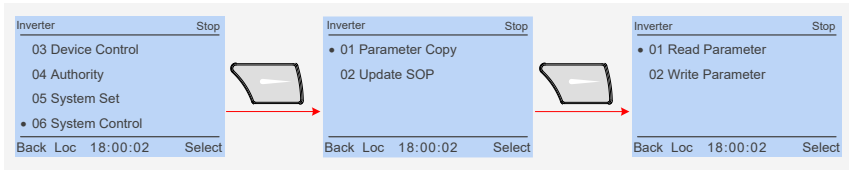
Table 4-8 Equipment operating description

Task	Action
Backup/edit and store equipment parameter	Open "menu"-"system setting"-"parameter copy", select "read parameter" or "write parameter". Make sure you have already logged in the system before this operation.
Program update	Open "menu"-"system setting"-"panel update" Make sure you have already logged in the system before this operation.

4.7.1 Parameter Copy

You can use the parameter copy function to transfer the parameter settings of one device to the other device. That is to copy the parameter settings of one device, download and copy them to other devices that needs the same parameter settings. The device that reads the parameter / write parameters is the device in which the current panel is operating, and the specific device information is displayed in the upper left corner of the interface.

Figure 4-10 Access the parameter copying interface after login



1) Read parameter

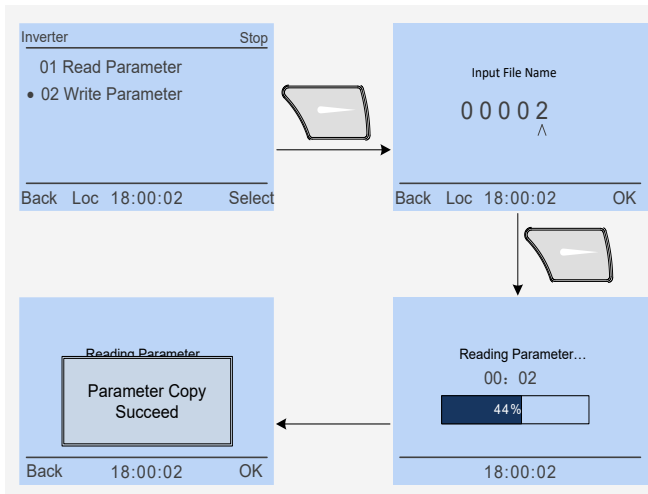
This operation shall read all parameters of the system.

Set the user-defined part of the file name before performing reading parameters. The user-defined part consists of 5 digits. use and to select the digit, and / to change the value (range: 0-9 and A-Z) of the digit. The format of the file name is as follows: Device name-5-digit user-defined part-time.ipv

For example: Inverter--00002-0302134905.ipv, Where inverter is the device name, 00002 is the user-defined part, 02 March, 13: 49: 05.

Press to read and copy the parameters after the file name is set. "Handle succeed" will display on the interface after reading and copying parameter is finished. Specific reason will be displayed if reading and copying parameter failed.

Figure 4-11 Read parameter



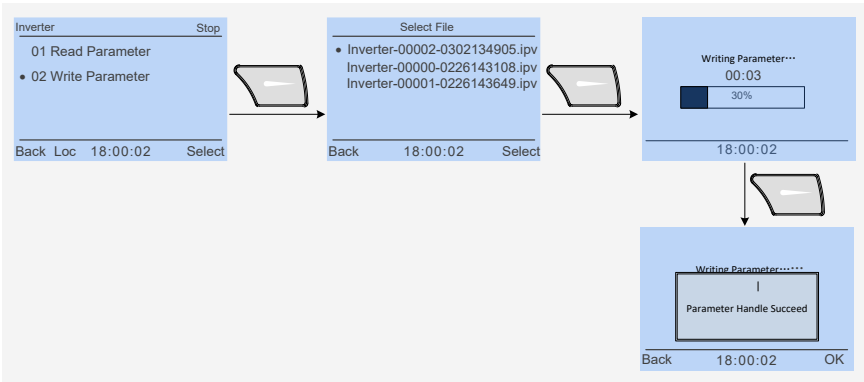
2) Write parameters

Different from reading parameters, there are two kinds of parameter write mode: "all copy mode" and "non-motor parameter copy mode". All copy mode copies all the parameters except the communication parameters. The non-motor parameter copy mode is tailored for parameters of certain parameter range. Please check the user manual of the device.

The procedures of write parameters is as follows:

- ① Press  to select the target device;
- ② Enter "write parameter" menu, then press  to select the already read parameter file;
- ③ In the copy mode interface, select the parameter copy mode you need;
- ④ Wait until the parameter download completes.

Figure 4-12 Example of write parameters



Note

All parameter files that are copied, saved and read are stored in SD card, ensure that the SD card is properly installed before using the parameter copy function.

Example: Copy all the parameters in 2# device to 3# device.

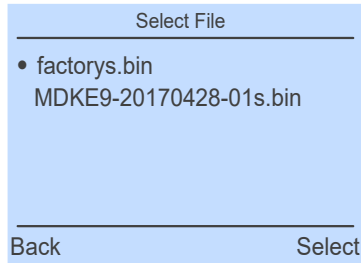
- ① Enter "key login" menu, enter the password to log into the system;
- ② Press  to select the 2# device;
- ③ Read the parameters in 2# device, and name it as "00002";
- ④ Press , select 3# device;
- ⑤ Write the "00002" parameter to 3# device via "all copy mode".

4.7.2 Update Device

Update procedure:

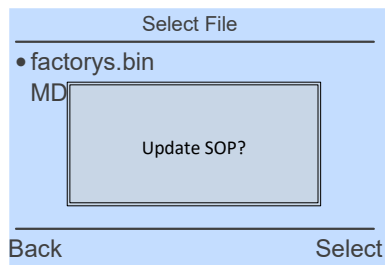
- ① Copy the update file from the PC to SD card under the SYS/Update path:
- ② Enter the panel upgrade menu after login. For details of panel login, see "4.5.1 Panel Login".
- ③ Select the files to be updated from the list.

Figure 4-14 Select the files to be updated



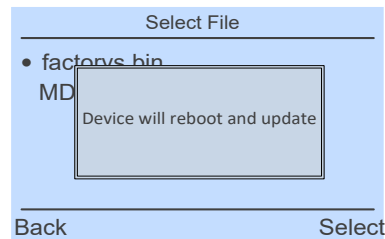
- ④ Press  to select at the pop-up window.

Figure 4-15 Confirming program update



- ⑤ Wait for the panel to complete restart, and the file burning procedure ends.

Figure 4-16 Example of program update processing



Note

Copy the update file from the PC to X: \SYS\PROGRAM. For details of the storage method of the SD card file, see "5.3 SD Card Replacement".

Chapter 5 Maintenance and Repair

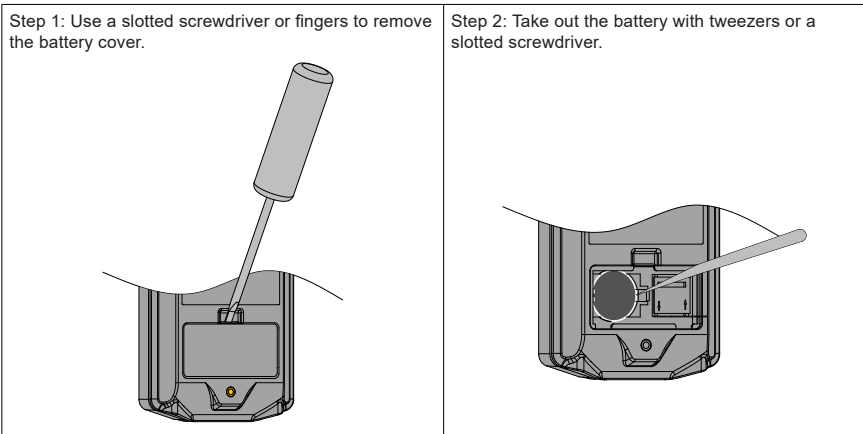
5.1 Daily and Periodical Inspection

Table 5-1 Carry out daily and periodic inspection as follows:

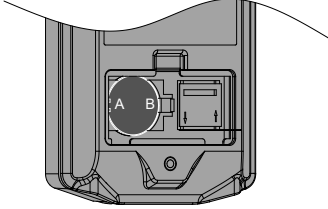
Type	Cycle	Inspection Items
Daily check	Daily	Check the ambient temperature, humidity, dust, foreign objects.
		Check whether there is abnormal noise
		Check whether the USB connector cover and the battery cover are installed properly.
		Check whether the LCD display is normal.
		Check whether the operation through the keys is normal.
		Check whether there is abnormal smell.
Periodical Inspection	1 year	Check whether the fastening parts get loosened.
		Check whether there is overheat.
		Check whether the USB connector gets damaged.
		Check whether the fastening parts of USB connector and RJ45 interface get loosened.
		Check whether the SD card is installed properly.
		Check whether the battery is installed properly.
Replace periodically	About 10 years	Replace the button battery.

5.2 Battery Replacement

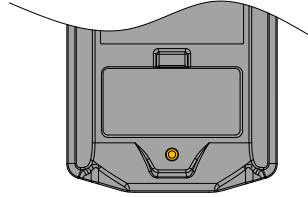
Figure 5-1 Replace the battery



Step 3: Place the battery into the clip (A side) first and then press other end (B side) of the battery.



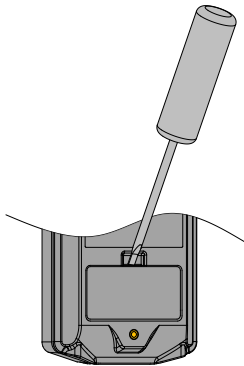
Step 4: Put the battery cover back



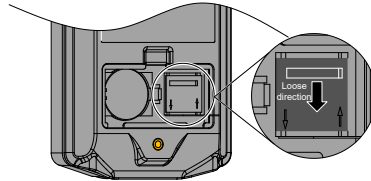
5.3 SD Card Replacement

Figure 5-2 Replace the SD card

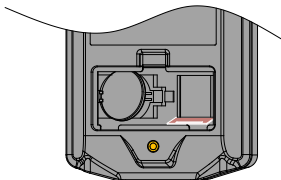
Step 1: Use a slotted screwdriver or fingers to remove the battery cover.



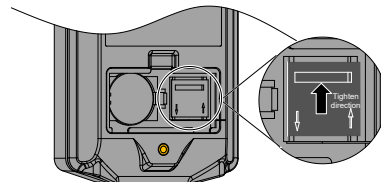
Step 2: Push the metal cover towards the unlock direction to release the SD card. Step 3: Lift the metal cover and place a new SD card.



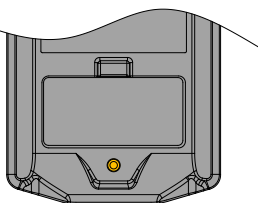
Step 3: Lift the metal cover and place a new SD card.



Step 4: Put down the metal cover and push the metal cover towards the lock direction to fix the SD card.



Step 5: Put the battery cover back.



Note

Plug and play is not supported. Replace the SD card when the MDKE9 is powered off.

Revision History

Date	Version	Change Description
October. 2017	A00	First issue.
April. 2019	A01	Updated the cover.
July. 2020	A02	Deleted the service hotline.
November 2020	A03	Made minor corrections.
April.2021	A04	Deleted section 4.7.2 "Device Burning". Updated the function definition of the "Loc/Rem" key in "2.4.1 Keys and Functions" and "4.1.3 Base Operation".

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